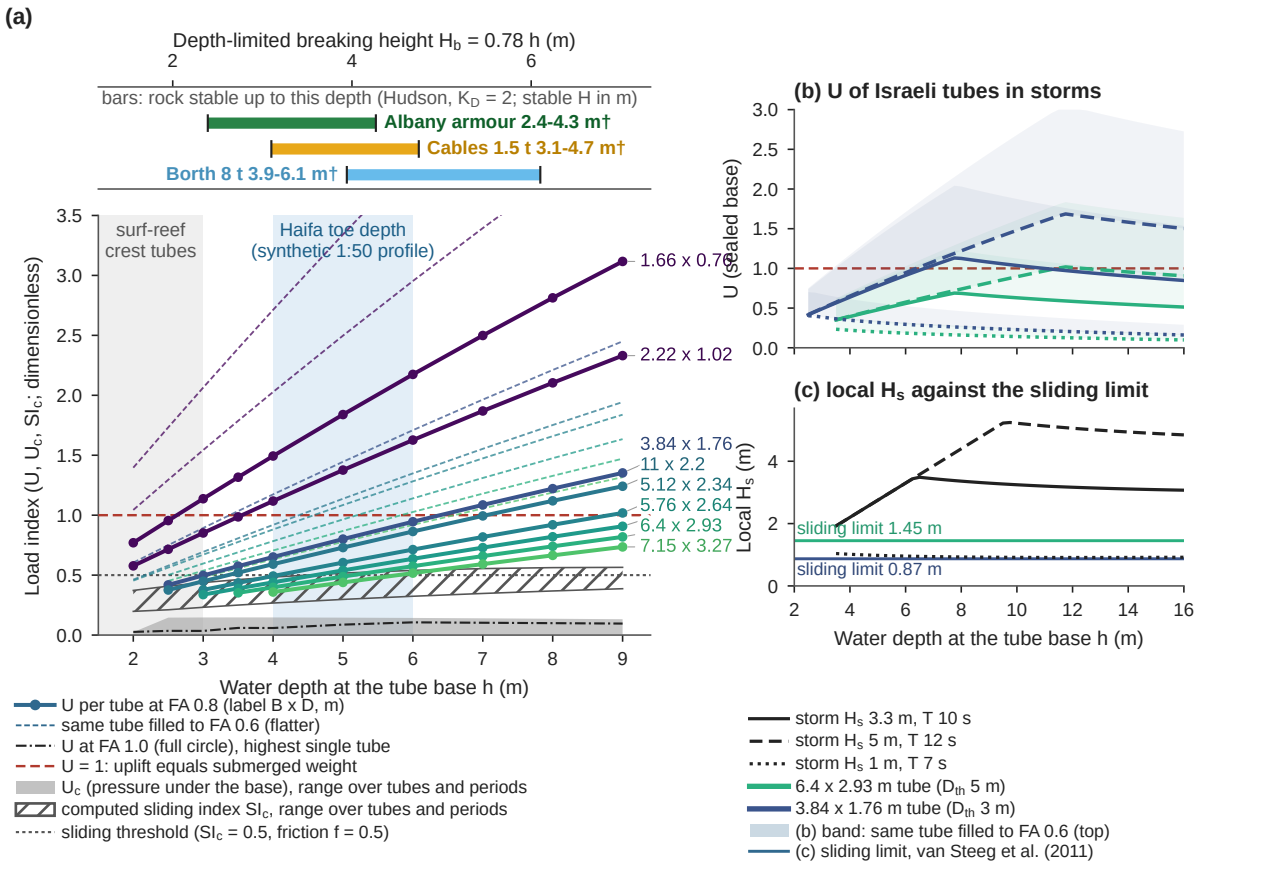

(a)
Depth-limited breaking height $H_b$ = 0.78 h (m)
2
4
6
bars: rock stable up to this depth (Hudson, $K_D$ = 2; stable H in m)
Albany armour 2.4-4.3 m†
Cables 1.5 t 3.1-4.7 m†
Borth 8 t 3.9-6.1 m†
surf-reef crest tubes
Haifa toe depth (synthetic 1:50 profile)
1.66 x 0.76
2.22 x 1.02
3.84 x 1.76
11 x 2.2
5.12 x 2.34
5.76 x 2.64
6.4 x 2.93
7.15 x 3.27
Load index (U, $U_c$, $SI_c$; dimensionless)
Water depth at the tube base h (m)
U per tube at FA 0.8 (label B x D, m)
same tube filled to FA 0.6 (flatter)
U at FA 1.0 (full circle), highest single tube
U = 1: uplift equals submerged weight
$U_c$ (pressure under the base), range over tubes and periods
computed sliding index $SI_c$, range over tubes and periods
sliding threshold ($SI_c$ = 0.5, friction f = 0.5)
(b) U of Israeli tubes in storms
U (scaled base)
(c) local $H_s$ against the sliding limit
Local $H_s$ (m)
sliding limit 1.45 m
sliding limit 0.87 m
Water depth at the tube base h (m)
storm $H_s$ 3.3 m, T 10 s
storm $H_s$ 5 m, T 12 s
storm $H_s$ 1 m, T 7 s
6.4 x 2.93 m tube ($D_{th}$ 5 m)
3.84 x 1.76 m tube ($D_{th}$ 3 m)
(b) band: same tube filled to FA 0.6 (top)
(c) sliding limit, van Steeg et al. (2011)